But there is another approach that, while more of a long shot, fills me with incomparable wonderment. In Chapter 11 we discussed how the effects of tiny quantum jitters can be seen in any clear night sky since they're stretched enormously by cosmic expansion, resulting in clumps that seed the formation of stars and galaxies. (Recall the analogy of tiny scribbles, drawn on a balloon, that are stretched across its surface when the balloon is inflated.) This realization demonstrably gives access to quantum physics through astronomical observations. Perhaps it can be pushed even further. Perhaps cosmic expansion can stretch the imprints of even shorter-scale processes or features—the physics of strings, or quantum gravity more generally, or the atomized structure of ultramicroscopic spacetime itself—and spread their influence, in some subtle but observable manner, across the heavens. Maybe, that is, the universe has already drawn out the microscopic fibers of the fabric of the cosmos and unfurled them clear across the sky, and all we need do is learn how to recognize the pattern.

Assessing cutting-edge proposals for deep physical laws may well require the ferocious might of particle accelerators able to re-create violent conditions unseen since moments after the big bang. But for me, there would be nothing more poetic, no outcome more graceful, no unification more complete, than for us to confirm our theories of the ultrasmall—our theories about the ultramicroscopic makeup of space, time, and matter—by turning our most powerful telescopes skyward and gazing silently at the stars.